

Data Preparation and Understanding, Data cleaning, Transforming





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- Data Preparation
- Data Profiling
- Data Cleaning
- Data Integration
- Data Transformation





Data Preparation

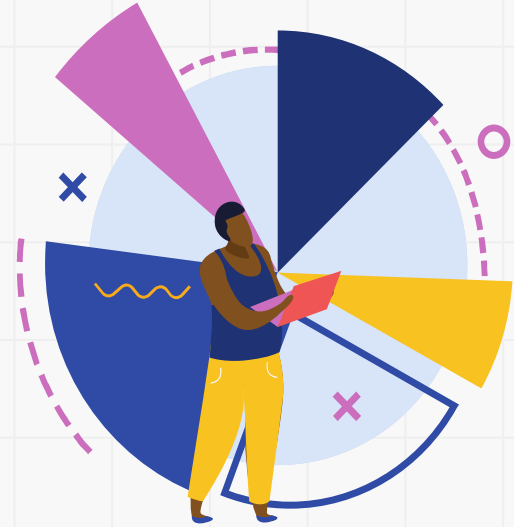
- Data preparation is the process of cleaning and transforming raw data prior to processing and analysis. It is an important step prior to processing and often involves reformatting data, making corrections to data, and combining datasets to enrich data.
- Data preparation is the process of gathering, combining, structuring and organizing data so it can be used in business intelligence (BI), analytics and data visualization applications.
- The components of data preparation include data preprocessing, profiling, cleansing, validation and transformation; it often also involves pulling together data from different internal systems and external sources.

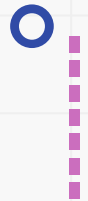


Why Data Preparation?

There are several reasons why we need to prepare the data:

- ❖ By preparing data, we prepare the miner so that when using prepared data, the miner produces better models faster.
- ❖ Good data is essential for producing efficient models of any type.
- ❖ Data should be formatted according to the required software tool.
- ❖ Data need to be made adequate for the given method.





Benefits of Data Preparation

Data preparation helps:

Fix errors quickly- Data preparation helps catch errors before processing. After data has been removed from its original source, these errors become more difficult to understand and correct.

Produce top-quality data- Cleaning and reformatting datasets ensures that all data used in analysis will be high quality.

Make better business decisions- higher quality data that can be processed and analyzed more quickly and efficiently leads to more timely, efficient and high quality business decisions.





Data Preparation Steps

01

Data Profiling

02

Data discretization

03

Data cleaning

04

Data integration

05

Data transformation

06

Data reduction

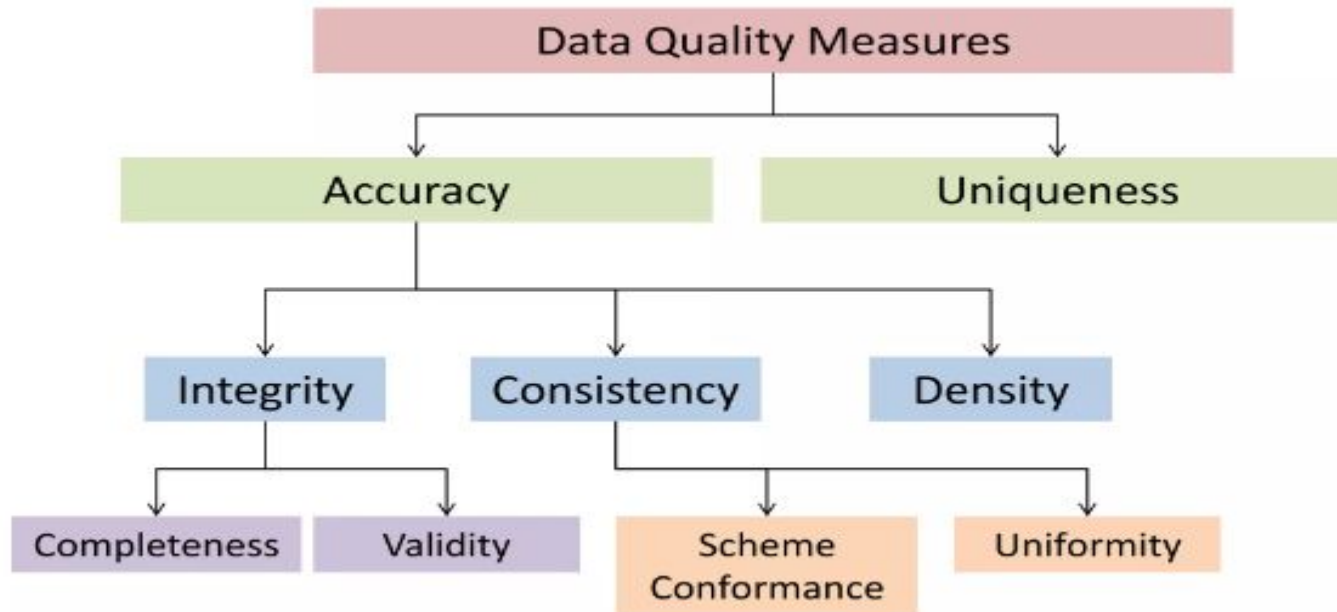
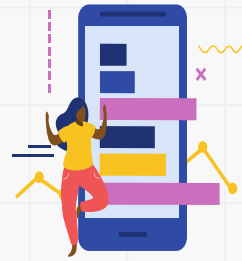


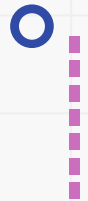
Data Profiling



Data Profiling

Data Profiling: Sourcing, selecting and auditing appropriate data





Data Quality: Why Preprocess the Data?

- Measures for data quality: A multidimensional view
 - Accuracy: correct or wrong, accurate or not
 - Completeness: not recorded, unavailable, ...
 - Consistency: some modified but some not, dangling, ...
 - Timeliness: timely update?
 - Believability: how trustable the data are correct?
 - Interpretability: how easily the data can be understood?

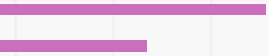




Major Tasks in Data Preprocessing

- **Data cleaning**
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data integration**
 - Integration of multiple databases, data cubes, or files
- **Data reduction**
 - Dimensionality reduction
 - Numerosity reduction
 - Data compression
- **Data transformation and data discretization**
 - Normalization
 - Concept hierarchy generation

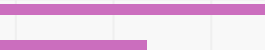




Data Preprocessing

- **Assuring** and **improving data quality** are two of the primary reasons for data preprocessing.
- There are common criteria to **measure** and **evaluate** the quality of data, which can be categorized into two main elements:
accuracy and **uniqueness**.

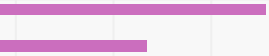




Data Profiling

- **Accuracy** is described as an aggregated value over the quality criteria: **Integrity, Consistency, and Density.**
- Intuitively this describes the extent to which the data are an **exact, uniform,** and **complete** representation of the mini-world: *the aspects of the world that the data describe.*

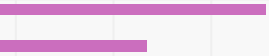




Data Profiling

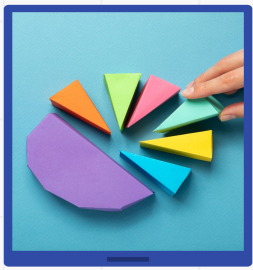
- **Integrity:** An integral data collection contains representations of all the entities in the mini- world and only of those.
- **Access data from any source** no matter the origin, format or narrative and integrating them together. Increased access to data means **less manual work, faster insights** and **faster time** to value realized by your organization.
- **Integrity** requires both **completeness** and **validity**.





Data Profiling

- **Completeness:** Complete data give a comprehensive representation of the mini-world and contain **no missing values**.
 - We achieve completeness within data cleansing by correcting anomalies and not just deleting them.
 - It is also possible that additional data are generated, representing existing entities that are currently unrepresented in the data.
 - A problem with assessing completeness is that *you don't know what you don't know*. As a result, there are no known gold standard data, which can be used as a reference to measure completeness.
- 
- 



Data Profiling

Validity: Data are valid when there are no constraints violated.

There are numerous mechanisms to increase validity including **mandatory fields, enforcing unique values, and data schema/structure.**



Data Profiling

Consistency: This quality concerns **syntactic anomalies** as well as **contradictions**.

The main challenge concerning data consistency is choosing which data source you trust for reliable agreement among data across different sources.

- ❖ **Schema conformance:** This is especially true for the relational database systems where the adherence of domain formats relies on the user.
- ❖ **Uniformity:** is directly related to irregularities.

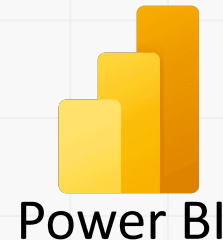


Data Preparation tools

Microsoft Power BI. This powerful tool makes it easy to deal with large volumes of data and then process it, creating insights automatically that will impact businesses. It also integrates seamlessly with other products in the Microsoft suite.

Tableau. Tableau makes data visualization a seamless activity, making it friendly for those who may not be as confident with the technical side of data. It also makes spotting trends within KPI's easy.

SAP Data Intelligence. Despite being a large box solution, SAP Data Intelligence is customizable to the organization's needs and KPI's.



Data Cleaning





Data Cleaning

Data cleaning is defined as the process of detecting and correcting (or removing) corrupt or inaccurate records from a record set, table, or database. Data cleansing can also be referred to as data cleansing, data scrubbing, or data reconciliation





Data Cleaning

Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error

- ❖ **incomplete**: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data

e.g., *Occupation*=" " (missing data)

- ❖ **noisy**: containing noise, errors, or outliers

e.g., *Salary*="−10" (an error)

- ❖ **inconsistent**: containing discrepancies in codes or names, e.g.,

Age="42", *Birthday*="03/07/2010"

Was rating "1, 2, 3", now rating "A, B, C"

discrepancy between duplicate records

- ❖ **Intentional** (e.g., *disguised missing data*)

Jan. 1 as everyone's birthday?

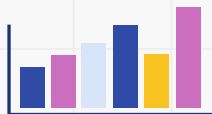


Data Cleaning

More precisely, the process of data cleaning could be explained as a four-stage process:

1

Define and identify errors in data such as incompleteness, incorrectness, inaccuracy, or irrelevancy.



2

Clean and rectify these errors by replacing, modifying, or deleting them



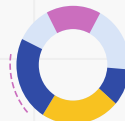
3

Document error instances and error types



4

Measure and verify to see whether the cleansing meets the user's specified tolerance limits in terms of cleanliness



Incomplete (Missing) Data

Data is not always available

E.g., many tuples have no recorded value for several attributes, such as customer income in sales data

Missing data may be due to

- Equipment malfunction
- Inconsistent with other recorded data and thus deleted
- Data not entered due to a misunderstanding
- Certain data may not be considered important at the time of entry
- Not register history or changes of the data

Missing data may need to be inferred

How to Handle Missing Data?

- Ignore the tuple: usually done when the class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably.
- Fill in the missing value manually: tedious + infeasible?
- Fill in it automatically with
 - a global constant : e.g., “unknown”, a new class?!
 - the attribute mean
 - the attribute mean for all samples belonging to the same class: smarter

most probable value: inference-based such as Bayesian formula or decision tree

Noisy Data

How to Handle Noisy Data?

- **Noise:** random error or variance in a measured variable
- Incorrect attribute values may be due to
 - faulty data collection instruments
 - data entry problems
 - data transmission problems
 - technology limitation
 - inconsistency in the naming convention
- **Other data problems** that require data cleaning
 - duplicate records
 - incomplete data
 - inconsistent data

- **Binning**
 - first sort data and partition it into (equal-frequency) bins
 - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- **Regression**
 - smooth by fitting the data into regression functions
- **Clustering**
 - detect and remove outliers
- **Combined computer and human inspection**
 - detect suspicious values and check by humans (e.g., deal with possible outliers)

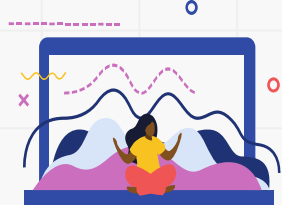
Data anomalies



- The term data **anomaly** describes any distortion of data resulting from the data collection process.

- From this perspective, anomalies include **duplication**, **inconsistency**, **missing values**, **outliers**, **noisy data** or any kind of distortion that can cause data imperfections.





Category	Percentage
Very important	100%
Important	100%
Not important	100%
Don't know	100%

01

02

02

03

03



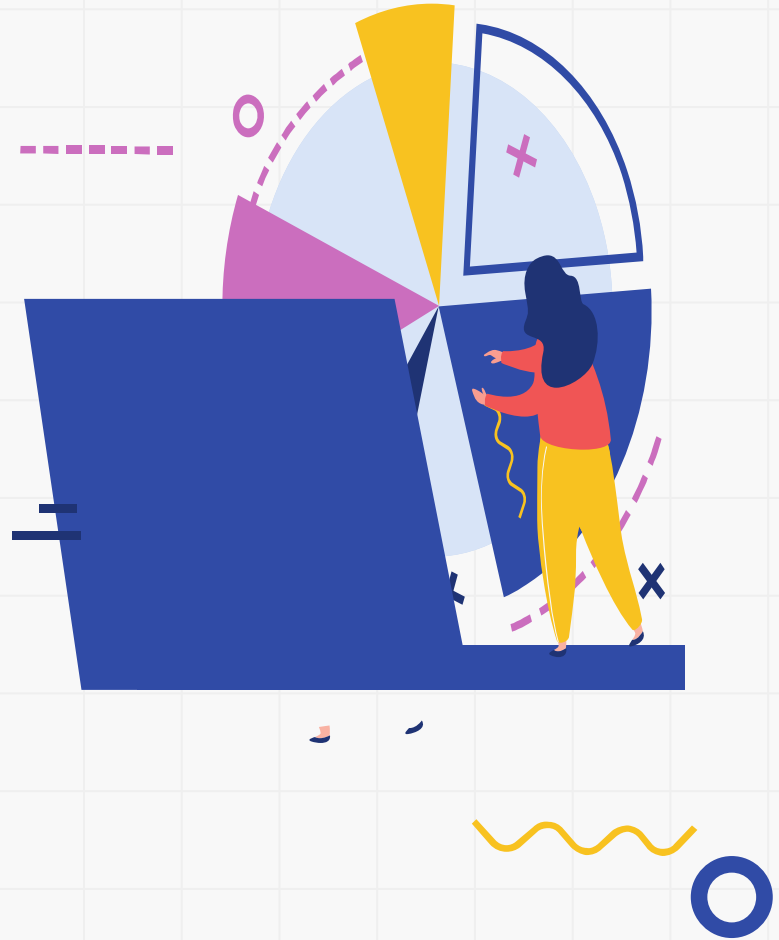
Data Cleaning Process



1. **Data Auditing:** This first step mainly identifies the types of anomalies that reduce data quality. Data auditing checks the data using validation rules that are pre-specified and then creates a report of the quality of the data and its problems. We often apply some statistical tests in this step for examining the data.
2. **Workflow specification:** The next step is to detect and eliminate anomalies by a sequence of operations on the data. The information collected from data auditing is then used to create a data-cleaning plan. It identifies the causes of the dirty data and plans steps to resolve them.
3. **Workflow execution:** The data cleaning plan is executed, applying a variety of methods on the data set.
4. **Post-processing and controlling:** The post-processing or control step involves examination of the workflow results and performs exception handling for the data mishandled by the workflow



Data Integration



Data Integration

- Existing data may be augmented through data enrichment. This commonly involves sourcing of additional information about the data points on which data are already held. For example, customer data might be enriched by obtaining socio-economic data about individual customers.
- Data integration is a crucial task in data preparation. Combining data from different sources is not trivial especially when dealing with large amounts of data and heterogeneous sources. Data are typically presented in different forms (structured, semi-structured or unstructured) as well as from different sources (web, database) that could be stored locally or distributed.



Data Integration

- **Data integration:**

Combines data from multiple sources into a coherent store

- Schema integration: e.g., A.cust-id , B.cust-#

Integrate metadata from different sources

- **Entity identification problem:**

Identify real world entities from multiple data sources, e.g., Bill Clinton = William Clinton

- Detecting and resolving data value conflicts

For the same real world entity, attribute values from different sources are different

Possible reasons: different representations, different scales, e.g., metric vs. British units



Data Transformation





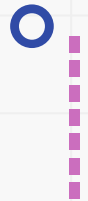
Data Transformation

A function that maps the entire set of values of a given attribute to a new set of replacement values s.t. each old value can be identified with one of the new values.

Methods:

- ❖ Smoothing: Remove noise from data
- ❖ Attribute/feature construction
 - New attributes constructed from the given ones
- ❖ Aggregation: Summarization, data cube construction
- ❖ Normalization: Scaled to fall within a smaller, specified range
 - min-max normalization
 - z-score normalization
 - normalization by decimal scaling
- ❖ Discretization: Concept hierarchy climbing





Data Transformation

It is frequently necessary to transform data from one representation to another. There are many reasons for changing representations:

- To generate symmetric distributions instead of the original skewed distributions:
- Transformation improves visualisation of data that might be tightly clustered relative to a few outliers.
- Data are transformed to achieve better interpretability.
- Transformations are often used to improve the compatibility of the data with assumptions underlying a modeling process, for example, to linearize (straighten) the relation between two variables whose relationship is non-linear. Some of the data mining algorithms require the relationship between data to be linear.





Discretization

- Discretization transforms continuous data into a discrete form. This is useful in many cases for: better data representation, data volume reduction, better data visualization and representing data at a various level of granularity for data analysis. Data discretization approaches are categorized as supervised, unsupervised, bottom-up or top down. Approaches for data discretization include Binning, Entropy based, Nominal to numeric, 3-4-5 rule and Concept hierarchy.





Discretization

Three types of attributes

Nominal—values from an unordered set, e.g., color, profession

Ordinal—values from an ordered set, e.g., military or academic rank

Numeric—real numbers, e.g., integer or real numbers

Discretization: Divide the range of a continuous attribute into intervals

Interval labels can then be used to replace actual data values

Reduce data size by discretization

Supervised vs. unsupervised

Split (top-down) vs. merge (bottom-up)

Discretization can be performed recursively on an attribute

Prepare for further analysis, e.g., classification



Data Discretization Methods

Typical methods: All the methods can be applied recursively

Binning

Top-down split, unsupervised

Histogram analysis

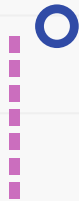
Top-down split, unsupervised

Clustering analysis (unsupervised, top-down split or bottom-up merge)

Decision-tree analysis (supervised, top-down split)

Correlation (e.g., x^2) analysis (unsupervised, bottom-up merge)





Concept Hierarchy Generation

- **Concept hierarchy** organizes concepts (i.e., attribute values) hierarchically and is usually associated with each dimension in a data warehouse.
- Concept hierarchies facilitate drilling and rolling in data warehouses to view data in multiple granularity
- Concept hierarchy formation: Recursively reduce the data by collecting and replacing low level concepts (such as numeric values for *age*) by higher level concepts (such as *youth*, *adult*, or *senior*)
- Concept hierarchies can be explicitly specified by domain experts and/or data warehouse designers
- Concept hierarchy can be automatically formed for both numeric and nominal data. For numeric data, use discretization methods shown.



Concept Hierarchy Generation for Nominal Data

Specification of a partial/total ordering of attributes explicitly at the schema level by users or experts

street < city < state < country

Specification of a hierarchy for a set of values by explicit data grouping

{Urbana, Champaign, Chicago} < Illinois

Specification of only a partial set of attributes

E.g., only street < city, not others

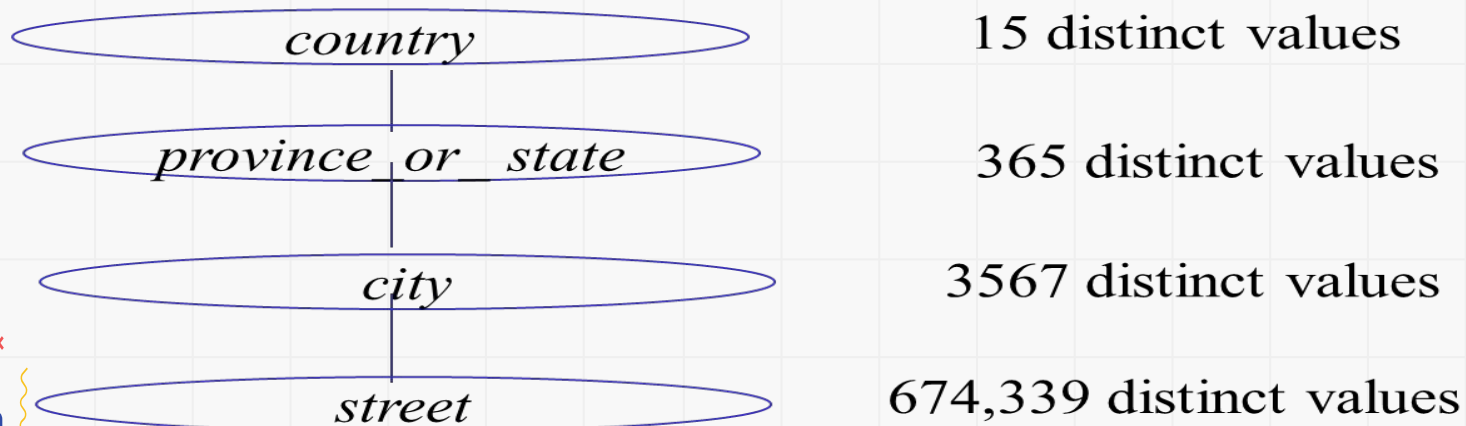
Automatic generation of hierarchies (or attribute levels) by the analysis of the number of distinct values

E.g., for a set of attributes: {street, city, state, country}

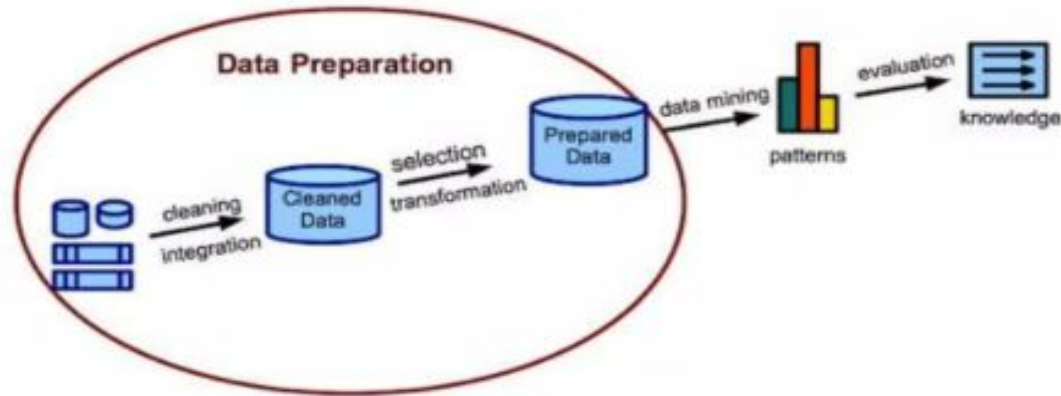


Automatic Concept Hierarchy Generation

- Some hierarchies can be automatically generated based on the analysis of the number of distinct values per attribute in the data set
 - The attribute with the most distinct values is placed at the lowest level of the hierarchy
 - Exceptions, e.g., weekday, month, quarter, year



Data Preparation Example



- There are multiple values that are commonly used to represent the virus. A virus like COVID-19 could be represented by 'SAR-Cov2', 'Corona', 'Covid' or 'Covid-19' to name a few.
- A data preparation tool could be used in this scenario to identify an incorrect number of unique values (in the case of virus, a unique count greater than suitable number in case of covid would raise a flag, as there are only few names aligned with virus). These values would then need to be standardized to use only an abbreviation or only full spelling in every row.

